



COURSE SYLLABUS
Bachelor of Science in Information Technology
First Semester, SY 2026-2027

COLLEGE	: College of Information and Communications Technology
PROGRAM	: Bachelor of Science in Information Technology
COURSE CODE	: IT 401W
COURSE TITLE	: Web Programming and Frameworks
CREDIT UNITS	: 1-unit Laboratory, 2-unit Lecture
PRE-REQUISITE	: IT 305W
FACULTY	: Dr. Aaron Paul M. Dela Rosa
CONSULTATION HOURS	: 3 hrs./week

COURSE DESCRIPTION:

This course introduces students to modern web application development using Python and the Django Web Framework. It covers foundational Python programming concepts, object-oriented programming, and progressively advances to web frameworks, MVC/MTV architecture, server-side rendering, database integration, and full CRUD (Create, Read, Update, Delete) operations. Students engage in hands-on development of dynamic, data-driven web applications, emphasizing clean code structure, security considerations, and framework-based best practices. The course culminates in a functional Django-based web application project, enabling students to integrate backend logic, templating, forms, and database management in a real-world development context. This course prepares learners for professional web development roles, further studies in full-stack development, and industry-relevant software engineering practices.

Bisyon

Ang Bulacan State University ay isang progresibong institusyong lumilikha ng kaalaman na kinikilala sa daigdig sa pinakamahusay na pagtuturo, nangungunang mga pananaliksik, at maagap na pagtugon sa pangangailangan ng komunidad.

Misyon

Ang Bulacan State University ay may hangaring humubogng mga propesyonal na mahuhusay. etikal, at handang maglingkod, na nakapag-aambag sa patuloy na pag-unladng bayan.

Core Values: ALAB BulSU!

Agap sa Pagbabago

Lingap sa Kapwa at Sangnilikha



Agapayan sa paggawa
Bayanihan sa banyuhay

The BulSU Ideal Graduate Attributes (BIG A) reflect the graduate’s capacity as:

- A. Makadiyos at Makatao (Ethical and Socially Responsible);
- B. Matalino at Malikhain (Critical and Creative Thinker);
- C. Mapanagutan at Maasahan (Accountable and Dependable Professional); and
- D. Makabansa at Makabuluhan ang Gawa (Service-Oriented and Nationalistic)

Program Educational Objectives (PEO)

Program Educational Objectives (PEO)	University Mission			
	BIG A-a	BIG A-b	BIG A-c	BIG A-d
PEO1. Apply knowledge and technical competencies in various specialization tracks of Information Technology aligned to the responsive region’s specific development needs.	✓		✓	✓
PEO2. Prepare graduates to address various user needs involved in the selection, development, application, integration, and management of computing technologies within an organization through critical thinking and problem solving, being creative, collaborating, and communication effectively.	✓		✓	✓
PEO3. Design, implement, and evaluate the local, regional, national, and global impact of information and communications technology through logical writing, making presentations, conducting collaborative research, performing innovative production, rendering extension services, establishing international linkages and partnerships.	✓	✓	✓	✓
PEO4. Immerse and expose in an actual environment in industry, engage in planning self-learning, and understand professional, ethical, legal, security, and social issue and responsibilities in the utilization of information technology towards the realization of sustainable development goals.	✓	✓		✓

Program Outcomes (PO)

On completion of the course, the student is expected to be able to do the following:

Program Outcomes (PO) (CMO No. 25, Series of 2015)	Program Educational Objectives			
	PEO1	PEO2	PEO3	PEO4
a. Apply knowledge of computing, science, and mathematics appropriate to the discipline of IT.	✓	✓		





b. Understand best practices and standards and their applications.		✓		
c. Analyze complex problems and identify and define the computing requirements appropriate to its solution.	✓	✓		✓
d. Identify and analyze user needs and take them into account in the selection, creation, evaluation, and administration of computer-based systems.			✓	✓
e. Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.	✓		✓	
f. Integrate IT-based solutions into the user environment effectively.	✓			
g. Apply knowledge through the use of current techniques, skills, tools, and practices necessary for the IT profession.	✓			
h. Function effectively as a member or a leader of a development team, recognizing the different roles within a team to accomplish a common goal.		✓		✓
i. Assist in the creation of an effective IT project plan.	✓	✓		
j. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.		✓		
k. Analyze the local and global impact of computing information technology on individuals, organizations, and society.			✓	
l. Understand professional, ethical, legal, security, and social issues and responsibilities in the utilization of IT.				✓
m. Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.		✓		✓

Course Outcomes and Relationship to Program Outcomes

Course Outcomes	Program Outcomes												
After completing this course, the student must be able to:	a	b	c	d	e	f	g	h	i	j	k	l	m
CO1. Apply Python programming fundamentals and object-oriented principles in developing backend web applications.	E	E	D	E				E					E
CO2. Analyze and design web applications using MVC and MTV architectural patterns.		E	E	E								E	





CO3. Develop dynamic web applications using Django views, templates, models, and forms.	E		D	D	I								E	
CO4. Implement secure and efficient database operations using Django ORM and CRUD techniques.	E		D	D						D				
CO5. Integrate modular programming, third-party libraries, and frameworks to solve web development problems.				D									D	E
CO6. Demonstrate teamwork, communication skills, and professionalism through project development and presentation.						D	D					E		

Note: (I) Introductory Course to an Outcome (E) Enabling Course to an Outcome (D) Demonstrative Course to an Outcome

Course Alignment with the United Nations Sustainable Development Goals

United Nations Sustainable Development Goals	
1. No Poverty	☐ Include topics on economic inequality, poverty alleviation, and inclusive economic growth.
2. Zero Hunger	☐ Discuss food security, sustainable agriculture, and nutrition systems.
3. Good Health and Well-Being	☐ Cover health promotion, mental well-being, and universal healthcare systems.
4. Quality Education	☒ Integrate equitable access to education, lifelong learning, and inclusive teaching practices.
5. Gender Equality	☐ Highlight issues of gender equity, women's empowerment, and reducing gender-based discrimination.
6. Clean Water and Sanitation	☐ Address water resource management, sanitation systems, and access to clean water.
7. Affordable and Clean Energy	☐ Explore renewable energy sources, energy efficiency, and sustainable energy solutions.
8. Decent Work and Economic Growth	☐ Focus on employment opportunities, workplace safety, and sustainable economic policies.
9. Industry, Innovation, and Infrastructure	☒ Discuss sustainable infrastructure, industrialization, and innovation.
10. Reduced Inequalities	☐ Analyze policies and practices that address income and opportunity disparities.
11. Sustainable Cities and Communities	☐ Examine urban planning, disaster resilience, and sustainable living solutions.
12. Responsible Consumption and Production	☐ Address waste reduction, sustainable consumption, and ethical production practices.
13. Climate Action	☐ Discuss mitigation and adaptation strategies for climate change.
14. Life Below Water	☐ Explore marine conservation, sustainable fisheries, and ocean health.
15. Life on Land	☐ Cover biodiversity conservation, reforestation, and land-use practices.





16. Peace, Justice, and Strong Institutions	☐ Promote social justice, conflict resolution, and strong governance.
17. Partnerships for the Goals	☐ Include topics on global collaboration, partnership-building, and cross-sectoral initiatives.

OTHER REQUIREMENTS AND ASSESSMENTS:

The student will be assessed at other times during the term by the following:

- Regular Class Attendance / Work Ethics
- Activities and Laboratories
- Active Class Participation / Performance
- Submission of Projects and Assignments
- Passing Examinations

GRADING SYSTEM:

Laboratory/Activities/Project	40%
Term Examinations	30%
Quizzes	20%
Attendance/Class Participation/Recitation	10%
TOTAL	100%

GRADE	EQUIVALENT
1.00	96.50-100
1.25	93.50-96.49
1.50	90.50-93.49
1.75	87.50-90.49
2.00	84.50-87.49
2.25	81.50-84.49
2.50	78.50-81.49
2.75	75.50-78.49
3.00	75.00-75.49
4.00	Conditional Passed
5.00	Failed
INC	Incomplete
D	Officially Dropped
UD	Unofficially dropped
FDA	Failure due to Absences

LEARNING

EPISODES:





Intended Learning Outcomes (Aligned to COs)	Topics	Week	Learning Activities	Suggested Assessment	Plan of Action in Case of Alternative Delivery Mode
Explain course requirements, policies, and expected outcomes (CO6)	Course Orientation	1	Syllabus discussion, LMS navigation, diagnostic activity	Diagnostic quiz (ungraded)	Recorded orientation, online syllabus quiz
Explain basic programming concepts and Python syntax (CO1)	Introduction to Programming with Python Setting Up the Environment	2	Lecture-demo, guided coding	MCQs, short coding drills	Self-paced videos, LMS quizzes
Apply data types, variables, operators, and basic I/O operations (CO1)	Introduction to Data Types Introduction to Variables Basic Input and Output Operations Basic Operators	3	Hands-on coding, code walkthroughs	Coding exercises, formative quiz	Asynchronous coding tasks
Analyze program flow using conditional statements (CO2)	Comparison Operators and Conditional Execution	4	Code tracing, problem-solving sessions	MCQs, logic-based coding	Recorded problem-solving demos
Implement iterative solutions using loops and lists (CO1, CO2)	Loops Lists and List Processing	5	Guided lab activities, mini-challenges	Mini-program, quiz	Virtual coding lab
Modularize programs using functions and handle runtime errors (CO1, CO3)	Introduction to Functions Function Parameters and Arguments Return Values and Lambda Functions	6	Lecture-demo, debugging workshop	Coding exercises, MCQs	Video tutorials, discussion forum



	Tuples and Dictionaries Exception and Error Handling				
Manipulate and analyze textual data using string methods (CO1, CO3)	Introduction to Strings String Slicing and Basic Operations String Methods	7	Text-processing labs, demonstrations	Coding exercises, quiz	Screencast lessons
Apply object-oriented programming principles to solve computing problems (CO1, CO3)	Introduction to Object-Oriented Programming Classes and Objects Attributes and Methods Principles of OOP Magic Methods	8	UML discussion, class-based coding, case analysis	OOP coding task, quiz	Recorded OOP walkthroughs
	MIDTERM EXAMINATION	9			
Apply modular programming and package management techniques (CO5)	Introduction to Modules Selected Python Modules Packages Python Package Installer	10	Lecture-demo, package installation	MCQs, short coding task	Installation demo videos
Analyze and compare MVC and MTV architectural frameworks (CO2)	Introduction to Software Architecture and Separation of Concerns	11	Concept mapping, framework comparison	MCQs, reflection activity	Video case studies



	Model-View-Controller (MVC) Framework Model-Template-View (MTV) Framework Comparing MVC and MTV				
Develop basic web applications using Django framework (CO3, CO5)	Introduction to Django Setting Up the Virtual Environment and Installing Django Creating a Django Project and Application Django Views and URL Mapping	12	Guided setup, hands-on Django lab	Coding task, quiz	Virtual lab environment
Implement data models and database operations using Django ORM (CO3, CO4)	Introduction to Django Models Database Migrations Django ORM and CRUD Operations Django Admin Interface	13	CRUD demonstrations, admin customization	Coding exercises, quiz	Screencast demos
Render dynamic content using Django templates (CO3)	Introduction to Django Templates Rendering Dynamic Content Template Tags and Filters	14	Template-based coding activities	Coding tasks, quiz	Asynchronous coding



	Template Inheritance and Reusability				
Process and validate user input securely using Django forms (CO4)	Introduction to HTML Forms and HTTP Methods Handling Forms in Django Views Django Forms and Validation CSRF Protection and Secure Form Handling	15	Form handling labs, guided practice	Coding exercises	Online consultations
Integrate full CRUD operations into a web application (CO3, CO4, CO5)	Overview of CRUD in Django Create Operation Read Operation Update Operation Delete Operation	16	Project development	Project milestone submission	Remote project mentoring
Present and defend a functional Django-based web application (CO6)	Project Presentation	17	System demonstration, oral defense	Project rubric-based evaluation	Recorded presentation submission
	FINAL EXAMINATION	18			

FINAL COURSE OUTPUT:

As a culminating requirement for IT 401W – Web Programming and Frameworks, students shall design, develop, and present a functional, database-driven web application using Python and the Django Web Framework.

The project must demonstrate the integration of Python programming fundamentals, object-oriented programming, Django framework components, database management, and full CRUD (Create, Read, Update, Delete) operations.





The final course output serves as concrete evidence of the students’ ability to apply theoretical concepts and technical skills to a real-world web development problem.

General Project Requirements:

Each student (or approved group) must submit and present a Django web application that includes the following:

- 1. Backend Logic
 - Implemented using Python and Django
 - Proper use of views, models, and URLs
 - Clean, readable, and modular code
- 2. Database Integration
 - Use of Django ORM
 - At least one main data model
 - Complete CRUD functionality
- 3. User Interface
 - Dynamic content rendered using Django templates
 - Use of template inheritance
 - Basic but functional UI layout
- 4. Forms and Validation
 - Use of Django Forms
 - Input validation and error handling
 - CSRF protection enabled
- 5. System Functionality
 - Data creation, viewing, updating, and deletion
 - Logical navigation and flow
 - No critical runtime errors
- 6. Documentation and Presentation
 - Brief system overview
 - Explanation of system features
 - Oral presentation and live system demonstration

RUBRICS FOR CHECKING:

Criteria	Excellent (90–100%)	Very Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)	Points
Application Functionality (25 pts)	All features work correctly; complete CRUD	Minor issues but CRUD works correctly	CRUD implemented but with noticeable errors	Incomplete or non-functional CRUD	25



	operations with no errors				
Use of Django Framework (20 pts)	Correct and effective use of models, views, templates, forms, and URLs	Mostly correct use with minor issues	Basic use of Django components	Poor or incorrect use of framework components	20
Database Design & ORM Usage (15 pts)	Well-designed models; efficient ORM queries	Acceptable model design; minor inefficiencies	Basic models with limited ORM usage	Poor or incorrect database implementation	15
Code Quality & Organization (15 pts)	Clean, readable, well-structured, properly commented code	Mostly readable with minor organization issues	Understandable but poorly organized code	Disorganized, unreadable, or error-prone code	15
User Interface & Usability (10 pts)	Clear layout, good navigation, dynamic content displayed correctly	Usable interface with minor layout issues	Basic interface with limited usability	Poor UI, confusing navigation	10
Security & Validation (5 pts)	Proper validation, CSRF protection implemented	Minor validation issues	Minimal validation	No validation or security measures	5
Presentation & Communication (10 pts)	Clear, confident explanation; strong technical justification	Clear explanation with minor gaps	Basic explanation of features	Poor explanation; unable to defend work	10
Total					100

REFERENCES:

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3. Greenfeld, A., & Roy, A. (2018). Two scoops of Django 2.x: Best practices for the Django web framework. Two Scoops Press.
4. Mozilla Developer Network. (2026). Web development documentation. <https://developer.mozilla.org>
5. Python Software Foundation. (2026). Python documentation. <https://docs.python.org>
6. Tobin-Hochstadt, S., & Van Horn, M. (2019). Programming languages: Principles and paradigms. MIT Press.
7. W3Schools. (2026). Python and web development tutorials. <https://www.w3schools.com>





Republic of the Philippines
City of Malolos, Bulacan
BULACAN STATE UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY



CLASS POLICIES:

1. The University rules and regulations will be observed.
2. Each student is expected to be in attendance at every class period.
3. You are responsible for what you missed. No special exams/activities/assignments, except if you missed a major exam. Assignment/requirement due dates must strictly be followed. Late requirements will no longer be accepted. Submit a written explanation signed by the BSIT Program Chair if you miss a major exam.
4. Eighteen (18) hours of allowable absences/tardiness. Exceeding before midterm is UD (Unofficially Dropped); after midterm, it is 5.0.
5. Submit on time and make sure to follow all instructions thoroughly. If not, do not submit; that is worth nothing.
6. Academic Dishonesty in any form is 5.0. All forms of academic dishonesty will be treated severely and swiftly. The copying and submission of another's work as your own constitutes cheating and/or plagiarism. There is no room for Academic Dishonesty in this class or any other.
7. Any form of copying or plagiarizing from past or current students' project code or report will result in disciplinary action and an automatic FAIL grade for the entire group or student(s) involved.
8. In case of illness, emergencies, etc., students are required to give an excuse letter signed by their respective guardians. A valid ID must also be presented for verification of the signature.

Prepared by:

(sgd.) Aaron Paul M. Dela Rosa, DIT

Checked by:

(sgd.) Edwin S. Garcia, MSIT
BSIT Program Chair

Approved by:

(sgd.) Digna S. Evale, DIT
Dean



Declaration

I have read and understood the above syllabus in full and in participating in this course I agree to the above rules. I have a clear understanding of the policies and my responsibilities, and I have discussed everything unclear to me with the professor.

I will adhere to the academic integrity and policy and I will treat my fellow students and my professor with due respect.

I understand that this syllabus can be modified or overruled by announcements of the professor in class or on any social media site at any time

Student's Printed name

Signature

Date

Student's Copy

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Instructor's Copy

